

CS & IT ENGINEERING



THEORY OF COMPUTATION

REGULAR EXPRESSION

Lecture No.- 02



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Recap of Previous Lecture



Topic

Regular Expressions
??????



Topics to be Covered



Topic

Regular Expression Properties

Topic

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$F.A \longleftrightarrow \text{Regular Expression}$

Topic

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Topic

??



Topic : Regular Expression

$L = \{a^n b^n \mid n \geq 0\}$ not Possible

- ① The simplest way of representing a regular language is known as Regular expression.
- ② For every regular language regular expression can be constructed.
- ③ To construct regular expression following 3 operators are used.
 - $+$ is known as union operator
 - $.$ is known as concatenation operator
 - $*$ is known as Kleene closure operator



Topic : NOTE

- For one regular language many number of regular expressions can be possible.
- One regular expression can generate only one regular language.

#Q. How many states are there in minimal DFA that accept following regular expression.

$$(1) \quad \underline{(a+b)} \underline{(a+b)} \underline{(a+b)} \underline{(a+b)} \xrightarrow[\text{(n+2)}]{\text{exactly}} 6 \text{ States}$$

$$(2) \quad \underline{(a+b+\epsilon)} \underline{(a+b+\epsilon)} \underline{(a+b+\epsilon)} \xrightarrow[\text{(n+2)}]{\text{at most}} 5 \text{ states}$$

$$(3) \quad (a+b)(a+b)(a+b)(a+b)^* \xrightarrow{\text{at least 3}} 4$$

$$(4) \quad \left[(a+b)^4 \right]^* \xrightarrow{\text{Divisible by 4}} \textcircled{4} \underline{\text{States}}$$

$\{a, b, aa, bb, \dots\}$

#Q. Construct regular expression that generates set of all strings of a's and b's where each string starting and ending with same symbol.

$$a(a+b)^*b + b(a+b)^*a$$

$$a + b + a(a+b)^*a + b(a+b)^*b$$

#Q. Construct regular expression that generates set of all strings of a's and b's where having substring aab.

#Q. Construct regular expression that generates set of all strings of a's and b's where ^xhaving substring aba(or) bab.

each string

$$\underline{(a+b)^*} (aba+bab) \underline{(a+b)^*}$$

#Q. Construct regular expression that generates set of all strings of a's and b's where 4th input symbol is a from left side.

$$(a+b)(a+b)(a+b)a(a+b)^*$$

#Q.) Construct regular expression that generates set of all strings of a's and b's where 4th input symbol is b from end. (R.H.S)

$$\left\{ \underbrace{(a+b)^*}_{\text{min}} \underbrace{b}_4 \underbrace{(a+b)}_3 \underbrace{(a+b)}_2 \underbrace{(a+b)}_1 \right\} \quad \left\{ 2^4 = \underline{16} \text{ states} \right\}$$

$\downarrow$
min DFA ?

(2^n)

#Q. Construct regular expression that generates set of all odd length palindrome strings over $\{a, b\}$.

Not Possible

#Q. Construct regular expression that generates set of all odd length palindrome strings of English language.



Topic : NOTE

- Palindrome languages over more than one symbol are not regular .Hence regular expression not possible.
- Palindrome languages over one symbol are regular.

$$\{\emptyset = 0\}$$

$$\{\epsilon = 1\}$$

$$(1) \quad R + \emptyset = \emptyset + R = R$$

$$(2) \quad R \cdot \emptyset = \emptyset \cdot R = \emptyset$$

$$(3) \quad R + \epsilon = \epsilon + R \neq R$$

$$(4) \quad R \cdot \epsilon = \epsilon \cdot R = R$$

$$(5) \quad (R^*)^* = (R^*)^+ = (R^+)^* = R^*$$

$$(6) \quad R \cdot R^* = R^+ = R^* R$$

$$(7) \quad \epsilon^* = \epsilon$$

$$(8) \quad \epsilon^+ = \epsilon$$

$$(9) \quad \emptyset^* = \epsilon$$

$$(10) \quad \emptyset^+ = \emptyset$$

$$(11) \quad \overbrace{(a + b)^*}^{ba} \neq \underline{a^*} \underline{b^*}$$

$$(12) \quad (a + b)^* \neq (a \cdot b)^* \quad \underline{ababab}$$

$$(13) \quad a(\underline{ba})^* = (ab)^* a$$

$$(14) \quad R^+ + \epsilon = R^* = R R^* + \epsilon$$

$$\begin{aligned}
 (14) \quad & \underline{(a^* b^*)^*} \\
 & (a + b)^* = (a + b^*)^* \\
 & = (a^* + b^*)^* \\
 & = (a^* + b)^* \\
 & \quad \quad \quad \searrow \\
 & \quad \quad \quad = (a^* b^*)^*
 \end{aligned}$$

$$(15) \quad \textcircled{a^*} + \textcircled{a^*} = a^* = a^* a^*$$

$$(16) \quad \begin{matrix} \{a|b\} & \{a|b\} \\ a + b = b + a \end{matrix}$$

$$(17) \quad \underline{a \cdot b} \neq \underline{ba}$$

$$(18) \quad (a+b)^* \neq a^* + b^*$$

[MCQ]

#Q. Identify language accepted by following regular expression

$$b^*(a^* \cdot \phi \cdot b + \underline{ab} + a\phi^*b^*)(b + \phi)^*$$

$$b^* \left[\overset{\phi}{a}b + ab^* \right] b^*$$

Exactly one a

$$b^* \left[a (b + b^*) \right] b^*$$

A

B

At least one a

$$b^* a b^* \cup b^*$$

C

At most one a

$$\{ \underline{b^* a b^*} \}$$

D

None

[MCQ]

#Q. Which of the following regular expressions are equivalent?

$\{even, odd\}$

- I. $(00)^* (\epsilon + 0) \rightarrow all$
- II. $(00)^* \rightarrow even$
- III. $0^* \rightarrow all$
- IV. $0(00)^* \rightarrow odd$

A (I) And (II)

B (ii) and (iii)

C (i) And (iii)

D (iii) and (iv)

[MCQ]

#Q. Which of the following pair of regular expressions are not equal

- A** $(r^*)^*$ and $(r+)^*$
- B** $(r + \epsilon)^*$ and r^*
- C** $(\underline{rr} + \epsilon)^*$ and r^*
- D** None of the above

(C)

THANK - YOU